APM348 Project Outline

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Title: Content-Centric Modeling of Rage-Bait Lifecycles on Social Media with Toxicity Feedback
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1. Background and Motivation

This project studies how platform ranking can drive a tradeoff between short-run engagement and long-run platform health. Prior evidence shows engagement-based algorithms can amplify divisive content [4, 5], while toxic content can affect user behavior [6]. Moral-emotional content also spreads more readily [3]. We model this using a compartmental ODE system where **content items** (not users) are the primary dynamical units.

2. Previous Mathematical Models

SIR-style models and rumor models [2] give threshold and stability tools for diffusion, and recent work applies these ideas to social-media spread [7]. For example, SVFR-type models focus on users' decisions and local network behavior. Toxicity-focused compartment models also exist [8, 9]. The main gap is that most models compartmentalize *users*; our model compartmentalizes *content lifecycle states* and couples them to toxicity and user attrition, describing a more global scale.

3. Proposed Model and Planned Contributions

State variables are I, V, F, S, τ, U for ignored, viral, fatigued, moderated/suppressed content, toxicity, and active users. Core transitions are:

$$\dot{I} = \frac{\rho U}{1+U} - \beta_{\text{eff}} IV - (\delta + \mu_c)I, \quad \dot{V} = \beta_{\text{eff}} IV - \gamma_{\text{eff}} V - \mu_c V, \quad \dot{F} = \gamma_{\text{eff}} V - \mu_c F, \quad \dot{S} = \delta I - \mu_c S,$$

$$\dot{\tau} = \phi V - \psi \tau, \qquad \dot{U} = \nu - \lambda_u (1 + w\tau) U,$$

with

$$\beta_{\text{eff}} = \alpha \beta_0 (1 + \kappa \tau), \qquad \gamma_{\text{eff}} = \gamma_0 (1 + \eta \tau).$$

Here ρ is content injection per user, μ_c natural content decay, and δ moderation/suppression. Main contribution: a content-centric model that links amplification, toxicity, and user-base dynamics in one system. For simplicity, direct moderation is modeled only from I via δI; an extension can include moderation from V or F. Natural decay μ_c represents loss of visibility/attention (content leaving the system), while S tracks explicit moderation removals. For the empirical Higgs fit, we use a reduced IVF calibration model for the retweet cascade with a decaying exogenous seed term $\lambda(t) = \lambda_0 e^{-d_\lambda t}$. This improves the data fit for the observed tail without changing the main IVFS policy model written above.

4. Planned Analysis and Simulations

- **Calibration data:** SNAP Higgs activity stream [1]. Verified counts: 563,069 total events, 354,930 retweets, peak at hour 78 with 30,602 RTs.

- **Parameter fitting:** fit a reduced IVF calibration model to a 100-hour normalized retweet window (scale-invariant) using a bounded Powell search with warm starts. The fitted calibration parameters are $\beta_0, \gamma_0, \lambda_0, d_\lambda, V_0$. The full IVFS scenario parameters are then held fixed when we run the policy model. On the pressure side, the toxicity couplings ϕ, ψ are still not jointly identified from Higgs alone; instead, the current constrained anchor is a reference-constrained fit that uses Higgs replies for proxy shape and Jigsaw only to fix the external scale ratio ϕ/ψ. So the toxicity side should be interpreted as partially constrained rather than fully data-calibrated.

- **Benchmarking:** compare the reduced IVF helper model against a classical SIR fit on the same normalized Higgs window. In the current run, the IVF helper model gives the stronger spike-region fit ($R^2 \approx 0.889$ versus 0.690 for SIR), supporting the added lifecycle structure.

- **Analytical work:** derive equilibria, a disease-free threshold analogue $\mathcal{R}_0$, and local Jacobian stability conditions.

- **Numerical scenarios:** engagement-first ($\alpha = 0.9$), moderate ($\alpha = 0.5$), health-first ($\alpha = 0.2$), with long horizons (e.g., $t = 1000$) for steady values of V^*, τ^*, U^* and $E = \alpha V^* U^*$. Under the current calibration, the threshold occurs at $\alpha(\mathcal{R}_0 = 1) \approx 0.316$, so the health-first case is subcritical while the moderate and engagement-first cases remain above threshold.

- **Continuation sweep:** compute long-run $\tau^*(\alpha)$ for $\alpha \in [0.05, 1.0]$ and report as continuation evidence (not a formal bifurcation claim unless stability checks support it).

- **Sensitivity:** vary $\kappa, \eta, \phi, \psi, \lambda_u$ to quantify robustness, with particular attention to ϕ because toxicity is only partially identified. A dedicated ϕ-sweep now serves as a core robustness check: with ψ fixed, the current model value $\phi = 0.056$ is compared against the larger external Jigsaw-implied value $\phi \approx 0.146$. The main qualitative ordering remains stable across this range: engagement-first yields the highest long-run toxicity, moderate remains in the middle, and the health-first case stays suppressed because it is already below the calibrated $\mathcal{R}_0 = 1$ threshold. On the tau side, the current constrained anchor is the *reference-constrained* fit: the external reference fixes $r = \phi/\psi$, and then ψ is chosen along the line $\phi = r\psi$ to best match the Higgs reply proxy. In the current run this gives approximately $\phi \approx 0.417$, $\psi \approx 0.286$, with constrained proxy-fit quality $R^2 \approx 0.41$.

- **Uncertainty quantification:** parametric Poisson bootstrap ($B = 500$) gives 95% CIs for $\beta_0 \in [0.9982, 1.0019]$, $\gamma_0 \in [0.2964, 0.3040]$, and $\alpha(\mathcal{R}_0 = 1) \in [0.3129, 0.3199]$. Local profile-likelihood curves around the optimum show clear minima for both β_0 and γ_0, supporting practical identifiability under the current calibration model.

- **Additional spread validation:** refitting the reduced IVF helper model on three representative 48-hour active windows from the same Higgs event record gives local R^2 values between about 0.69 and 0.95. The fitted parameters shift across those shorter windows, so the correct interpretation is local burst-decay robustness rather than globally stable event parameters.

5. Expected Results and Conclusion Plan

Current results indicate that higher amplification improves short-run virality but worsens long-run toxicity and user retention. They also suggest a meaningful threshold effect: once α drops below the calibrated $\mathcal{R}_0 = 1$ boundary, viral activity is no longer self-sustaining. The new ϕ-robustness and reference-constrained tau analysis do not remove the toxicity-identification limitation, but they do show that the main policy ranking is not an artifact of one arbitrary pressure setting. The additional 48-hour active-window fits likewise suggest that the reduced IVF helper model captures burst-decay behavior across multiple active phases of the same event record, even though the shorter-window parameters are not constant. The conclusion will summarize equilibrium and stability findings, compare policy scenarios, discuss implications for platform moderation and ranking policy, and state limitations clearly.

Limitations. Homogeneous mixing, deterministic dynamics, and single-index toxicity are the main structural simplifications. The fitted β_0, γ_0 and calibration-side seed parameters are estimated from a single main Higgs window, so the empirical spread fit is stronger than the toxicity fit. Additional 48-hour active-window refits support the helper model's local burst-decay shape, but they do not imply globally stable parameters across the whole record. In particular, toxicity is still only partially identified from the same dataset: the reference-constrained fit gives the best (ϕ, ψ) pair consistent with the external scale, but the Jigsaw comparison remains an external reference rather than a direct platform match. The current ϕ-sweep and reference-constrained tau fit should therefore be read as robustness and constrained-calibration arguments, not as substitutes for true same-dataset toxicity labels. Even so, the bootstrap intervals are narrow and the local profile curves show well-defined minima for both β_0 and γ_0, lending practical identifiability support for the spread component. Extensions to network-aware and stochastic models are natural next steps.

References

[1] Stanford Network Analysis Project. Higgs Twitter Dataset. `https://snap.stanford.edu/data/higgs-twitter.html`, 2013.

[2] D. J. Daley and D. G. Kendall. Epidemics and rumours. *Nature*, 204:1118, 1964.

[3] W. J. Brady, J. A. Wills, J. T. Jost, J. A. Tucker, and J. J. Van Bavel. Emotion shapes the diffusion of moralized content in social networks. *Proc. Natl. Acad. Sci. USA*, 114(28):7313–7318, 2017.

[4] F. Huszár, S. I. Ktena, C. O'Brien, L. Belli, A. Schlaikjer, and M. Edizel. Algorithmic amplification of politics on Twitter. *Proc. Natl. Acad. Sci. USA*, 119(1):e2025334119, 2022.

[5] S. Milli, M. Carroll, Y. Wang, S. Pandey, S. Zhao, and A. Dragan. Engagement, user satisfaction, and the amplification of divisive content on social media. PNAS Nexus, 4(3):pgaf062, 2025.

[6] G. Beknazar-Yuzbashev, R. Jiménez Durán, J. McCrosky, and M. Stalinski. Toxic content and user engagement on social media: Evidence from a field experiment. CESifo Working Paper 11644, 2025.

[7] Y. Li and L. Shao. Using an epidemiological model to explore the interplay between sharing and advertising in viral videos. *Scientific Reports*, 14:11351, 2024.

[8] M. Maleki, V. Mead, and N. Agarwal. Applying an epidemiological model to evaluate the propagation of toxicity on Twitter. In *Proc. HICSS-55*, 2022.

[9] E. Addai, N. Agarwal, and N. Yousefi. Modeling and simulation of interventions' effect on the spread of toxicity in social media. *Online Social Networks and Media*, 46:100309, 2025.

[10] Y. Jing, H. Wang, and Z. Zhang. Improved SIR advertising spreading model and its effectiveness in social network. *Procedia Computer Science*, 129:215–218, 2018.

[11] M. De Domenico, A. Lima, P. Mougel, and M. Musolesi. The anatomy of a scientific rumor. *Scientific Reports*, 3:2980, 2013.

[12] D. Borkan, L. Dixon, J. Sorensen, N. Thain, and L. Vasserman. Nuanced metrics for measuring unintended bias with real data for text classification. In *Proc. WWW Companion*, 2019.